

Element I

Jake Steinberg, Astrid Santiago, Caleb Bridgeman, Anne-Marie Georgi, Jack Hanson

1) Importance of Test (Testing Plans)

1. (High Priority) Safe

- a. Members of the group have tried and experimented with this device throughout the design and construction process. It is important to try this device and eliminate every possible way that would result in injury.

2. (High priority) User- friendly

- a. Similar to the safety feature, we experimented with this design for countless uses to ensure that it is completely safe and user-friendly. It is difficult to get injured while using this device.

3. (High priority) Precise

- a. We used the laws of physics and understood that the negative pressure acts as a vacuum to make the water rise. Every time the device is used, the water has to rise due to the negative pressure.

4. (High priority) Repeatability

- a. The device went through countless uses and gave the same data for each user, marking it repeatable.

5. (High Priority) Inexpensive

- a. The device is affordable, and it does not require a lot of money to operate. Its total cost is \$21.59.

Comprehension of Testing Procedure

RECAP: For our testing, we're using a syringe to act as a person inhaling through our MIP device. The device has a one-way filter valve, a funnel, and a clear PVC pipe with a crescendo cut that controls how much water moves up when you inhale. There's also a slit at the bottom of the PVC that lets water in when it's placed in water. By twisting the pipe, we can change how much water flows through.

The whole idea is based on fluid dynamics and pressure differences. Pulling back on the syringe creates negative pressure, which pulls water up—just like how inhaling creates suction in your lungs. The one-way valve ensures air only moves in one direction, similar to how breathing works. The crescendo cut lets us control water flow, like how airway resistance changes how much air you can breathe. By testing different angles of the PVC pipe with the syringe and seeing how the water reacts, we'll get a better idea of how well our design works and what we might need to adjust. This connects to how air and fluids move, so our testing method is backed by actual science.

Detail of Design Analysis

The way this device works is as the inhaling increases, the water rises. We opened a google document, put an image of a ruler, and printed the image out. We cut the picture of the ruler and taped it to the clear pvc pipe. As the water reaches higher numbers, the stronger the inhale. There is a little piece of styrofoam in the tube that does two things: makes it nearly impossible to inhale water and to show how high the water goes.

Summary of Data Implications

What is next for this project is calibrating and finding exact data. Right now, the water symbolizes only the strength of the inhalation. The data is also ambiguous because of the strength and resistance of the inhalation. We are struggling but dealing with competing factors. The variables are not 100% understood yet.